Xing Zhou

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RESEARCH INTERESTS

Limnologist, oceanographer, and modeler with research interests in model development and questions related to regional-scale ecosystem dynamics and biogeochemical cycling.

EDUCATION

- **Ph.D. in Atmospheric Science**
  Michigan Technological University 2017 –2023
  Dissertation title: *Using Hydrodynamic and Biophysical Models to Study the Mechanisms Mediating Algal Blooms in the Great Lakes*
  Advisor: *Dr. Pengfei Xue*
- **Bachelor of Atmospheric Science**
  Lanzhou University 2013-2017

ACADEMIC EMPLOYMENTS

- **Postdoctoral Fellow**
  Georgia Institute of Technology 2023-Present
  Advisor: *Dr. Annalisa Bracco*
- **Graduate Research Assistant**
  Michigan Technological University 2017-2023

TEACHING & MENTORING

Teaching

- Teaching Assistant & Co-Lecturer, *Climate Models (EAS 8803)*, Georgia Tech, 2024
- Co-Lecturer, *Introduction to Oceanography (EAS 4300)*, Georgia Tech, 2025; Guest Lecture 2024

Mentoring

- Gabe Lobdell (Undergraduate, University of Georgia, Summer REU, 2025)
- Mireya Ramirez (Undergraduate, Georgia Tech, 2024, PURA Award)
- Aya Kanawati (Undergraduate, Georgia Tech, Spring 2024)

GRANTS & FUNDING

- Co-PI – *Modeling the Dispersal and Connectivity of Marine Larvae with GenAI Agents*. Co-funded by Microsoft and two GeorgiaTech research institutes (the Brook Byers Institute for Sustainable Systems and the Institute for Data Engineering and Science). $50,000 + $50,000 Microsoft Azure credit (09/2024–07/2025). (*Served as Co-PI in accordance with Georgia Tech policy that generally does not allow postdoctoral researchers to serve as lead PI, but I conceived, initiated, and led the project.*). PI: A. Bracco; Co-PIs: X. Zhou, R. Wu, J. Abernethy.

PUBLICATIONS

Published

1) **Zhou, X.,** Xiao, S., Ramirez, M., & Bracco, A. (2025). Modeling river and urban related microplastic pollution off the southern United States. *npj Emerging Contaminants*, *1*(1), 9.
2) Xue, P., Huang, C., Zhong, Y., Notaro, M., Kayastha, M. B., **Zhou, X.**, ... & Kemp, E. (2025). Enhancing winter climate simulations of the Great Lakes: insights from a new coupled lake–

ice–atmosphere (CLIAv1) system on the importance of integrating 3D hydrodynamics with a regional climate model. *Geoscientific Model Development*, 18(13), 4293-4316.

3) Gardner, S. T., Rowe, M. D., Xue, P., **Zhou, X**., Alsip, P. J., Bunnell, D. B., ... H¨o¨ok, T. O. (2024). Climate-influenced phenology of larval fish transport in a large lake. *Limnology and Oceanography Letters.*
4) **Zhou, X**., Lopera, L., Roa-Var´on, A., Bracco, A. (2024). Modeling the larval dispersal and connectivity of Red Snapper (*Lutjanus campechanus*) in the Northern Gulf of Mexico. *Progress in Oceanography*, 224, 103265.
5) **Zhou, X**.,Chaffin, J. D., Bratton, J. F., Verhamme, E. M., Xue, P. (2023). Forecasting microcystin concentrations in Lake Erie using a Eulerian tracer model. *Journal of Great Lakes Research*, 49(5), 1029-1044.
6) **Zhou, X.,** Rowe, M.D., Liu, Q., Xue, P. (2023). Comparison of Eulerian and Lagrangian transport models for harmful algal bloom forecast in Lake Erie. *Environmental Modelling & Software*, 162, 105641.
7) **Zhou, X.,** Auer, M. T., Xue, P. (2021). Open lake phosphorus forcing of *Cladophora* growth: Modeling the dual challenge in great lakes trophic state management. *Water*, 13(19), 2680.
8) Chaffin, J. D., Bratton, J. F., Verhamme, E. M., Bair, H. B., Beecher, A. A., Binding, C. E., ... **Zhou, X**. (2021). The Lake Erie HABs Grab: A binational collaboration to characterize the western basin cyanobacterial harmful algal blooms at an unprecedented high-resolution spatial scale. *Harmful Algae*, 108, 102080.
9) Xue, P., Schwab, D. J., **Zhou, X**., Huang, C., Kibler, R., Ye, X. (2018). A hybrid Lagrangian-Eulerian particle model for ecosystem simulation. *Journal of Marine Science and Engineering*, 6(4), 109.

In revision/ Under review

1) **Zhou, X.,** †Novi, L., †Hay, M. E., Montoya, J. P., Aliu, A., Realff, M. J., & Bracco, A. Changing Drivers of the Great Atlantic Sargassum Belt: From Physical Forcing to Ecological Control. *Nature Communications* (accepted). (†Co–first authors)
2) **Zhou, X.,** Xue, P., Rowe, M. D., Alsip P. J., ... Gardner S. T. Modeling Study of Phytoplankton Responses in Lake Michigan to a Changing Climate. *Limnology and Oceanography*. (under review)
3) **Zhou, X.,** Bracco, A., Ito, T., Reinhard, C. T. Numerical assessment of hypothetical large-scale river-based alkalinity modification scenarios in the northern Gulf of Mexico. *Earth's future* (under review). Preprint available at https://cdrxiv.org/preprint/408

In preparation

1) **Zhou, X.,** Wang, G., Wu, R., & Bracco, A. SWARM: An open-source generative AI agent-based framework for modeling marine larval dispersal and settlement. (In preparation for *Science Advance*).

PRESENTATIONS (presenter underlined)

1) **Zhou, X. (2025).** Modeling Approaches for Sustainable Lake Ecosystems: From Climate Change Impacts to AI-Augmented Frameworks. *Uppsala University Seminar, Sweden*. March 11. (Invited talk)
2) **Zhou, X.,** Bracco, A., Ito, T., & Reinhard, C. T. (2025). Numerical assessment of hypothetical large-scale river-based alkalinity modification scenarios in the Northern Gulf Coast. *AGU Fall Meeting 2025*, New Orleans, LA, USA. Dec 15-19. (Poster)
3) **Zhou, X.,** Wang, G., Wu, R., Abernethy, J., & Bracco, A. (2025). IchthyopAgent: A generative AI-agent based Lagrangian tool for modeling ichthyoplankton dynamics: Development and application. *Ocean AI Workshop, Halifax Convention Centre, Halifax, Canada.* Nov 4-5. (Talk)
4) Bracco, A., Lopera, L., **Zhou, X**., & Herrera, S. Ecosystem Connectivity in the Northern Gulf: From Corals to Fish. *Benthic Ecology Meeting 2025*, Mobile, AL, April 1–4. (Talk)

5) **<u>Zhou, X</u>**., Wu, R., Wang, G., Bracco, A., Abernethy, J. (2025). Modeling the Dispersal and Connectivity of Marine Larvae with GenAI Agents. *ASLO 2025 Aquatic Science Meeting*, Charlotte, NC, Mar 26-31. (Talk)
6) <u>Ramirez, M</u>., **Zhou, X**., Xiao, S., Bracco, A. (2025). Investigating the Fate of Microplastic Pollution in the Northern Gulf of Mexico and its impact on Marine Habitats. *ASLO 2025 Aquatic Science Meeting*, Charlotte, NC, Mar 26-31. (Poster)
7) **<u>Zhou, X</u>**., Bracco, A., Ito, T., Reinhard, C. (2024) The Impact of Submesoscale Processes on Carbon Dynamics in the Gulf of Mexico during Large-Scale Alkalinity Modification. *AGU fall meeting*, Washington, D.C, Dec 09-13, 2024. (Poster)
8) **<u>Zhou, X</u>**., Xue, P., Rowe, M. D., Alsip P. J., ... Gardner S. T. (2024). Impacts of Climate Change on Phytoplankton Dynamics in Lake Michigan: a Biophysical Modeling Study. *ASLO*, Madison, WI, Jun 2-7,2024. (Talk)
9) **<u>Zhou, X</u>**., Bracco, A., Lopera, L., & Roa-Var´on, A. (2024). Red Snapper (*Lutjanus Campechanus*) Larval Dispersal and Connectivity in the Northern Gulf of Mexico: A High-Resolution Modeling Study. *Ocean science meeting 2024*, New Orleans, Louisiana, Feb 18-23,2024, (Talk)
10) **<u>Zhou, X</u>**., Xue, P., Rowe, M. D., Alsip P. J., ... Gardner S. T. (2022). Impact of mixing regime shift on phytoplankton, and invasive mussel in Lake Michigan under a warming climate: a biophysical modeling study. *AGU fall meeting*, Chicago, Illinois, Dec 12-16, 2022. (Talk)
11) **<u>Zhou, X</u>**., Xue, P., Chaffin, J., Bratton, J., & Verhamme, E. (2022). Incorporation of microcystin production improves lake erie cyanobacterial bloom toxin forecasts. *In Joint Aquatic Sciences Meeting 2022*. Grand Rapids, Michigan, May 14-20,2022. (Talk)
12) **<u>Zhou, X</u>**., Xue, P., Liu, Q., & Rowe, M. D. (2021). Intercomparison of three HAB transport models in short-term forecasting of Lake Erie CHABs event. *In 26th Biennial CERF Conference*, Online, 1–4 and 8–11 November 2021. (Talk)
13) **<u>Zhou, X.</u>**, Xue, P., & Auer, M. T. (2020). Offshore P-forcing of *Cladophora* growth in the Lake Michigan nearshore: a one-dimensional modeling approach. *International Association for Great Lakes Research*, Online, June 9–11, 2020. (Poster)
14) <u>Xue, P</u>., Ye, X., **<u>Zhou, X</u>**., & Huang, C. (2018). A hybrid Lagrangian-Eulerian particle model for ecosystem simulation in Sandusky Bay. *Estuaries and Coastal Modeling*, Seattle, Washington, June 25-28, 2018. (Talk)
15) <u>Kibler, R</u>., Huang, C., **<u>Zhou, X</u>**., & Xue, P. (2018). Using a property-carrying particle model for ecosystem simulation: A case study of Sandusky Bay. *International Association for Great Lakes Research*, Toronto, Canada. June 18-22, 2018. (Talk)

HONORS AND AWARDS

- Judith Curry Award, Georgia Institute of Technology 2024
- Dean's award for outstanding scholarship, Michigan Technological University 2023
- GLRC Student Research Grant, Michigan Technological University 2020
- Visiting student in Texas Tech University awarded by China Scholarship Council 2015

PROFESSIONAL SERVICE

- **Editorial Board Member:** *npj Emerging Contaminants*
- **Proposal Reviewer:** NASA Research Initiation Awards (RIA24)
- **Journal Reviewer (selected):** *Progress in Oceanography*, *Communications Earth & Environment*, *Journal of Hydrology*, *Environmental Science & Technology, Scientific Report*s
- **Session Convener & Co-chair:** "Leveraging Biophysical Models to Understand and Mitigate Climate Change Impacts on Aquatic Ecosystems (SS20)," ASLO 2025 Aquatic Sciences Meeting

Outreach & Community Engagement

- Co-hosted interview with Freedom Middle School Robotics Program students on marine biology applications (2024)

- Served as Postdoctoral Representative in faculty hiring processes, School of Earth and Atmospheric Sciences, Georgia Institute of Technology (2024)

MEDIA RECOGNITION

- NCCOS news: Gulf of Mexico Red Snapper Larval Model Suggests Managing Fishery as Two Stocks (https://coastalscience.noaa.gov/news/gulf-of-mexico-red-snapper-larval-model-suggests-managing-fishery-as-two-stocks/)
- CMCC news: Research reveals how microplastics threaten Gulf of Mexico marine life (https://www.cmcc.it/article/research-reveals-how-microplastics-threaten-gulf-of-mexico-marine-life)